

Notes: Ben-David, Bhattacharya, Huang, and Jacobsen (JF, 2026)

The (Missing) Relation between Acquisition Announcement Returns and Value Creation

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1 Big picture

This paper asks whether acquisition announcement returns, especially cumulative abnormal returns around the deal announcement (CAR), are a reliable market-based measure of expected value creation in mergers and acquisitions. The standard event-study interpretation is that if an acquisition creates value for the acquirer, the acquirer’s stock price should rise around the announcement; if the deal destroys value, the acquirer’s stock price should fall. Ben-David, Bhattacharya, Huang, and Jacobsen (2026) challenges this convention directly.

The paper’s central result is stark: **announcement returns do not meaningfully predict commonly used or newly constructed ex post acquisition outcomes**. The authors show that CAR is largely uncorrelated with goodwill nonimpairment, short-term abnormal ROA, long-term abnormal ROA, and, except in a statistically positive but economically small way, deal completion. A simple characteristics-based model using information known at the announcement date predicts these outcomes far better than CAR .

The paper then asks why CAR fails. The proposed explanation is that acquisition announcements reveal two things at once:

$$\$CAR = \underbrace{NPV}_{\text{deal-related value}} + \underbrace{X}_{\text{stand-alone acquirer-related information}}.$$

Because acquisition decisions are endogenous, the announcement reveals information about the trigger for the acquisition and about the acquirer itself, not only about the transaction. The authors argue empirically that the acquirer-related component X dominates the deal-related component NPV , making it difficult to extract deal value from CAR .

2 Research question

The paper focuses on three linked questions.

1. **Measurement question:** Does acquirer CAR around an acquisition announcement measure the expected value created by the acquisition?
2. **Prediction question:** Is CAR at least correlated with ex post outcomes that should reflect value creation, such as nonimpairment, operating performance, completion, or long-run abnormal returns?
3. **Mechanism question:** If CAR fails, is the failure driven mainly by anticipation, completion uncertainty, selection, feedback, or by the fact that acquisition announcements reveal acquirer-specific information unrelated to deal NPV ?

3 Incremental contribution

The incremental contribution is not simply that CAR is noisy. Prior work already discusses anticipation, leakage, completion uncertainty, price pressure, serial acquisition effects, and target/acquirer bargaining complications. This paper’s contribution is to show that these concerns are not marginal measurement frictions but are severe enough to undermine the standard use of CAR as a value-creation proxy.

- **Comprehensive test of the CAR -equals- NPV convention.** The paper examines more than 47,000 U.S. acquisition announcements from 1980 to 2018, rather than relying on small samples or one outcome metric.
- **New transaction-level failure measure.** The authors manually construct transaction-specific goodwill impairment data after SFAS 142. This is important because goodwill impairment is tied to the target and therefore gives a transaction-level signal of acquisition failure.
- **Multiple outcome proxies.** The paper uses nonimpairment, short-term abnormal ROA, long-term abnormal ROA, deal completion, and long-run DGTW-adjusted buy-and-hold returns. These outcomes are positively correlated with each other, while their correlations with CAR are near zero.
- **Out-of-sample benchmark.** A simple model using deal and acquirer characteristics known at the announcement date predicts outcomes out-of-sample. CAR does not predict outcomes and does not even correlate with the predictable component of outcomes.
- **Systematic search for a “golden subset.”** The paper examines decades, industries, simple subsamples, and brute-force combinations of deal and acquirer characteristics. It fails to find a robust subset in which CAR consistently predicts outcomes.
- **Economic diagnosis of the failure.** The paper studies the dollar value implied by CAR and shows that $\$CAR$ varies much more with acquirer size than with deal size. This indicates that the acquirer-related signal swamps deal-related value information.

4 Core intuition

The standard interpretation of CAR relies on a clean-event logic: the acquisition announcement is the event, and the price reaction is the market’s estimate of the deal’s value to the acquirer. The problem is that acquisition announcements are not random news shocks. Managers choose when and what to acquire in response to information and strategic triggers that investors may not know.

For example, an acquisition could reveal that an internal R&D project failed, that the firm is entering a new market, that management believes the firm is undervalued, that it plans to abandon an existing strategy, or that the CEO has particular skill or overconfidence. Even if the deal itself has an NPV , the announcement also updates investors’ beliefs about the stand-alone acquirer. Thus CAR is a combined signal.

The authors argue that this combined signal is dominated by acquirer information. If CAR were mainly deal NPV , the dollar magnitude of CAR should scale with deal size. Instead, the variance and magnitude of $\$CAR$ scale far more strongly with acquirer market capitalization.

5 Data and sample construction

The acquisition sample comes from SDC and covers U.S. public acquirers from 1980 through 2018. The main sample includes acquisitions in which the acquirer is public, the target transaction value is available, the acquirer seeks at least 50% of the target, the acquirer owns less than 50% before the announcement, and the deal is not a leveraged buyout, spinoff, repurchase, self-tender, recapitalization, privatization, stake purchase, or acquisition of a partial/remaining interest.

The initial sample contains 47,543 deals. Of these, 42,354 are completed, 2,227 are withdrawn, and 2,962 are neither completed nor withdrawn or have unknown outcome status. The authors use different subsamples depending on the outcome variable because goodwill impairment, ROA, completion, and returns have different data requirements.

5.1 Main outcome measures

- **Goodwill nonimpairment.** A transaction-level dummy equal to one if the deal’s goodwill is not materially impaired within five years after completion. This is the paper’s new outcome measure. It is available primarily after the implementation of SFAS 142.
- **Short-term abnormal ROA.** The residual from a regression of postacquisition industry-adjusted ROA over years $t + 1$ to $t + 3$ on preacquisition industry-adjusted ROA.
- **Long-term abnormal ROA.** Similar to short-term abnormal ROA but measured over years $t + 4$ to $t + 6$, intended to capture longer-horizon synergy realization.
- **Completion.** A dummy equal to one if the announced deal is completed rather than withdrawn. This tests whether managers “listen” to the market reaction.
- **Long-run returns.** Five-year DGTW-adjusted buy-and-hold abnormal returns, used to validate whether predicted outcome measures contain economically meaningful information.

5.2 Announcement returns

The paper estimates abnormal returns using a market model with CRSP value-weighted returns. The main measures are:

- $CAR[-1, 1]$, the three-day announcement-window return;
- $CAR[-5, 5]$, an 11-day announcement-window return;
- $DealCAR[A - 2, C + 2]$, returns from two days before announcement to two days after deal completion.

6 Empirical design

The paper does not attempt to estimate a causal treatment effect of acquisitions on outcomes. Instead, it performs a measurement-validity test: if CAR measures expected acquisition value creation, then it should be positively related to future value-creation outcomes and to the predictable component of those outcomes.

The empirical approach has five layers.

1. **Visual bin tests:** sort transactions into CAR bins and plot realized outcomes.
2. **In-sample regressions:** regress outcome variables on CAR alone, then on CAR plus deal/acquirer characteristics and fixed effects.
3. **Subsample tests:** repeat the core regressions by decade, industry, acquirer type, target type, payment form, deal size, and other characteristics.

4. **Out-of-sample prediction:** fit *CAR*-only and characteristics-only prediction models in an early period and test predictive power in a later period.
5. **Economic decomposition of $\$CAR$:** ask whether the dollar value implied by *CAR* scales with deal size or acquirer size.

7 Main findings

7.1 Finding 1: *CAR* has almost no relation with ex post outcomes

The central empirical fact is that *CAR* is not meaningfully correlated with nonimpairment, short-term abnormal ROA, or long-term abnormal ROA. The relation with completion is positive but economically small. This matters because completion is not value creation; it only indicates whether the deal proceeds.

The visual evidence in Figure 2 shows nearly flat relationships between *CAR* bins and three of the four outcomes. The regression evidence in Table II confirms that *CAR* adds almost no explanatory power once standard deal and firm characteristics are included.

7.2 Finding 2: Characteristics predict outcomes much better than *CAR*

A simple characteristics-only model using information available at announcement predicts acquisition outcomes out-of-sample. These characteristics include acquirer market capitalization, leverage, free cash flow, Tobin's Q, previous-quarter market-adjusted returns, relative deal size, and dummies for stock-only payment, mixed payment, diversifying deals, competed deals, hostile deals, and public targets.

The important result is not only that characteristics predict outcomes. It is that *CAR* does not correlate with the characteristic-predicted component of outcomes. This means that *CAR* fails to reflect value-relevant information available to investors at the announcement date.

7.3 Finding 3: Characteristics-based predicted outcomes validate the ex post measures

The authors validate their outcome measures by linking predicted outcomes to long-run returns. When acquirers are sorted by characteristics-predicted outcomes, the top three deciles outperform the bottom three deciles by roughly 7.9% to 10.7% over five years. When acquirers are sorted by *CAR*-predicted outcomes, the return spreads are much smaller and often insignificant.

This supports the idea that the outcome proxies capture economically meaningful value creation, while *CAR* does not.

7.4 Finding 4: Anticipation, selection, truncation, and feedback are not enough

The authors test whether the failure of *CAR* can be explained by deal anticipation, uncertainty about completion, sample selection due to withdrawals, or managerial feedback. These forces exist, but they do not explain the core failure.

For example, extending the return window, excluding anticipated deals, focusing on high-completion-probability deals, and analyzing the benefits of listening to *CAR* do not rescue *CAR*. In the “listening” exercise, following *CAR* signals generates losses, while following the characteristics-based model generates large positive spreads.

7.5 Finding 5: *CAR*-based inferences about deal quality are unreliable

The paper shows that the types of deals that appear good under *CAR* do not align with the types of deals that perform well ex post. In particular, deal categories with high average *CAR* sometimes have the worst ex post outcomes. Table VI is especially striking: the group with the highest average *CAR* has poor nonimpairment and abnormal ROA outcomes.

7.6 Finding 6: *CAR* is dominated by acquirer-related information

The paper’s economic decomposition argues that $\$CAR$ contains both deal-related *NPV* and stand-alone acquirer information *X*. If $\$CAR$ were mostly *NPV*, its magnitude should scale with deal size. Instead, $\$CAR$ scales much more strongly with acquirer market capitalization.

The regression evidence in Table VII shows that acquirer size is roughly 6 to 13 times more important than deal size in explaining $\$CAR$ variation. This supports the paper’s conclusion that *CAR* is swamped by information about the acquirer rather than the transaction.

8 Interpretation and implications

The paper does not say that event studies are useless. It says that *CAR* around acquisition announcements is a poor measure of acquisition value creation. *CAR* remains a valid measure of the stock-market reaction to the announcement, but the market reaction is not the same as the deal’s *NPV*.

For empirical corporate finance, the implication is significant. Many papers classify acquisitions as value-creating or value-destroying based on *CAR*. This paper argues that those inferences should be reconsidered. Researchers should use richer outcome measures, characteristics-based models, transaction-level accounting outcomes, and long-run evidence rather than treating *CAR* as a sufficient statistic for deal value.

For teaching and practice, the paper is a caution against a common shortcut: a positive acquirer announcement return does not necessarily mean the acquisition creates value, and a negative return does not necessarily mean the deal destroys value. The announcement may reveal information about the acquirer that overwhelms the deal signal.

9 Potential limitations and useful caveats

- **Outcome measures are imperfect.** Goodwill impairment, abnormal ROA, completion, and long-run returns each have limitations. The authors address this by using multiple measures and showing that they correlate with each other and with long-run returns.

- **CAR may still matter for some questions.** CAR can be useful if the research question is about market reaction itself. The problem arises when CAR is interpreted as expected deal NPV.
- **The paper provides a diagnosis, not a perfect replacement.** The characteristics model performs better than CAR, but it is not a structural measure of deal NPV. The paper calls for more research on better ex ante acquisition-quality measures.
- **The acquirer-information component is difficult to observe.** The paper shows that it likely dominates CAR, but X itself is not directly observed. This is precisely why simply controlling for observable acquirer characteristics cannot solve the problem.

10 Tables and figures

10.1 Figure 1 — The distribution of CAR [-1, 1] with respect to relative size; Table I — Descriptive Statistics of the Measures

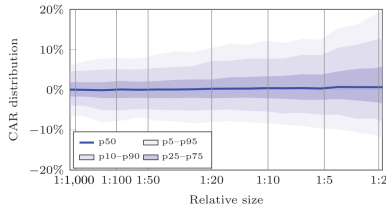


Figure 1. The distribution of CAR [-1, 1] with respect to relative size. This figure shows the distribution of cumulative abnormal returns (CAR [-1, 1]) around acquisition announcements, plotted against the ratio of transaction value to the acquirer's market capitalization, for deals with ratios below 1:2. CAR distributions vary only modestly across different deal sizes, indicating that CAR is only very weakly related to relative transaction size.
(Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 1 — The distribution of CAR [-1, 1] with respect to relative size

Table I
Descriptive Statistics of the Measures

This table reports descriptive statistics of the ex ante and ex post measures of acquisition quality. Panel A presents summary statistics, and Panel B shows correlations. All continuous variables are winsorized at the 1% level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A: Summary Statistics						
	N	Mean	Std Dev	P10	P50	P90
Nonimpairment	6,128	0.852	0.355	0	1	1
Short-term abnormal ROA (ST abROA)	28,710	-0.002	0.082	-0.078	-0.006	0.091
Long-term abnormal ROA (LT abROA)	22,577	-0.002	0.091	-0.086	-0.007	0.097
Completion	39,585	0.944	0.230	1	1	1
CAR[-1, 1]	28,710	0.009	0.066	-0.056	0.003	0.080
CAR[-5, 5]	28,710	0.008	0.100	-0.099	0.002	0.122
DealCARIA - 2, C + 2)	28,710	-0.011	0.199	-0.218	-0.001	0.183
DGFW-adjusted BHAR (Adj BHAR)	27,355	-0.096	1.225	-1.289	-0.293	1.214

Panel B: Correlations						
	Non-impair	ST abROA	LT abROA	Completion	CAR [-1, 1]	DealCAR [-5, 5]
Nonimpairment	1					
ST abROA	0.133***	1				
LT abROA	0.127***	0.671***	1			
Completion	—	—	—	1		
CAR[-1, 1]	-0.004	0.003	-0.010	0.015***	1	
CAR[-5, 5]	0.002	-0.002	-0.005	0.015***	0.628***	1
DealCARIA - 2, C + 2)	0.047***	0.020***	0.009	—	0.354***	0.398***
Adj BHAR	0.252***	0.221***	0.257***	0.018***	0.040***	0.070***

(b) Table I — Descriptive Statistics of the Measures

Reading notes. Figure 1 is the first clue that CAR is not behaving like deal NPV: the distribution of CAR varies only modestly with relative deal size. Table I documents the key descriptive fact: ex post outcomes are correlated with one another, while correlations between outcomes and CAR are close to zero.

10.2 Figure 2 — CAR [-1, 1] and ex post outcomes; Table II — Acquirer CAR and Acquisition Outcomes

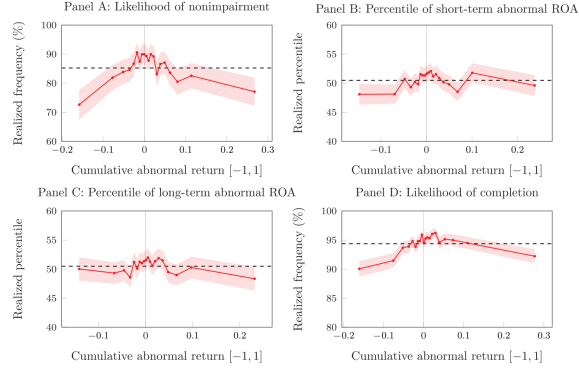


Figure 2. CAR [-1, 1] and ex post outcomes. Observations are sorted into 20 equally sized bins based on their CAR[-1, 1]. Panel A plots the percentage of transactions without impairment for each acquirer's CAR[-1, 1] vigintile (solid red line). Panels B and C present the average realized percentile of short- and long-term abnormal ROA, respectively, and Panel D presents the realized frequency of completion for each vigintile of CAR. The light red shading indicates 95% confidence intervals. The horizontal black dashed line represents the unconditional realized frequency (Panels A and D) and unconditional realized percentile (Panels B and C) in our sample. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 2 — CAR [-1, 1] and ex post outcomes

Table II
Acquirer CAR and Acquisition Outcomes

This table presents regression results analyzing the relationship between acquisition outcomes and acquirer cumulative abnormal returns (CAR). The dependent variables include a nonimpairment dummy (Panel A), short-term abnormal ROA (Panel B), long-term abnormal ROA (Panel C), and a completion dummy (Panel D). Columns (1) to (3) use CAR as the sole independent variable. Column (4) adds firm characteristics, and column (5) incorporates characteristics as well as year and industry fixed effects. Column (6) includes characteristics only, and column (7) combines characteristics with year and industry fixed effects. The control variables include log market capitalization, leverage and free cash flow scaled by lagged assets, Tobin's Q, prior-quarter market-adjusted stock returns, relative deal size, and dummy variables for stock-only, mixed-payment, diversifying, competed, hostile, and public target deals. Firm-level characteristics are measured in the year prior to the deal announcement. Standard errors are shown in parentheses, and ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

CAR Window:	[-1, 1]	[-5, 5]	[A - 2, C + 2]	[-1, 1]	n.a.		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Probability of Nonimpairment ($N = 6,128$, DV: Nonimpairment Dummy)							
CAR	-0.020 (0.105)	-0.008 (0.081)	0.089** (0.040)	0.082 (0.083)	0.080 (0.089)	Controls only	Controls only
Controls	—	—	—	Char	Year, Ind, Char	Char	Year, Ind, Char
Adjusted R^2	0.000	0.000	0.002	0.036	0.088	0.036	0.088
Panel B: Short-Term Abnormal ROA ($N = 28,710$, DV: Short-Term Abnormal ROA)							
CAR	0.004 (0.012)	-0.001 (0.006)	0.009** (0.003)	0.019 (0.011)	0.019* (0.010)	Controls only	Controls only
Controls	—	—	—	Char	Year, Ind, Char	Char	Year, Ind, Char
Adjusted R^2	0.000	0.000	0.000	0.026	0.078	0.026	0.078
Panel C: Long-Term Abnormal ROA ($N = 22,577$, DV: Long-Term Abnormal ROA)							
CAR	-0.015 (0.014)	-0.004 (0.007)	0.004 (0.003)	0.001 (0.012)	-0.002 (0.012)	Controls only	Controls only
Controls	—	—	—	Char	Year, Ind, Char	Char	Year, Ind, Char
Adjusted R^2	0.000	0.000	0.000	0.034	0.109	0.034	0.109
Panel D: Probability of Completion ($N = 39,585$, DV: Completion Dummy)							
CAR	0.048* (0.023)	0.031* (0.014)	—	0.050** (0.019)	0.041** (0.016)	Controls only	Controls only
Controls	—	—	—	Char	Year, Ind, Char	Char	Year, Ind, Char
Adjusted R^2	0.000	0.000	—	0.147	0.153	0.147	0.153

(b) Table II — Acquirer CAR and Acquisition Outcomes

Reading notes. Figure 2 bins deals by CAR and asks whether outcomes improve monotonically with the announcement return. They generally do not. Table II formalizes the result: CAR does not predict nonimpairment or abnormal ROA in any economically meaningful way. Completion is related to CAR , but the magnitude is very small.

10.3 Figure 3 — Acquirer CAR and acquisition outcomes: By decade; Table III — Out-of-Sample: Predicted versus Realized Outcomes

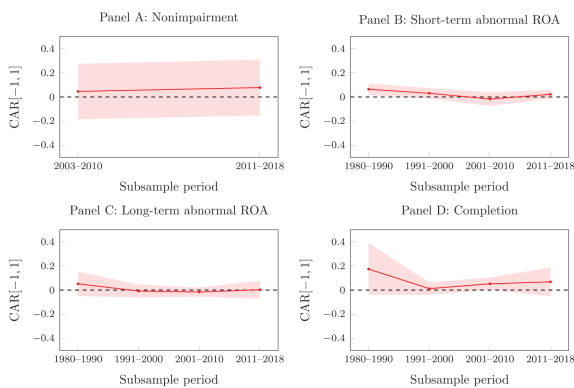


Figure 3. Acquirer CAR and acquisition outcomes: By decade. This figure plots coefficients and 95% confidence intervals for regressions of outcomes on CAR based on the specification in Table II, column (5), for each of the four decades in our sample (except for nonimpairment, which we can determine for only two decades due to data limitations). Panels A, B, C, and D use nonimpairment, short-term abnormal ROA, long-term abnormal ROA, and completion, respectively, as the key dependent variables. The red dots represent the point estimates, and the light red shading represents 95% confidence intervals.
(Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 3 — Acquirer CAR and acquisition outcomes: By decade

Table III
Out-of-Sample: Predicted versus Realized Outcomes

We first estimate OLS regressions of deal outcome measures on $CAR[-1, 1]$ only and characteristics only using the first half of transactions in each sample as a fit period to estimate coefficients. We then use the parameter estimates from this first half to predict outcomes in the second half of each sample. In Panel A, we assess the correlation between realized outcomes and predicted outcomes produced by the CAR-only model (columns (1), (3), (5), and (7)) and the characteristics-only model (columns (2), (4), (6), and (8)). In Panel B, we assess the correlation between the predicted outcome by the characteristics-only model and acquirer CAR. Standard errors are reported in parentheses below the coefficients. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A: Predicted versus Realized Outcomes								
Dependent Variable:	Realized Outcome							
	Nonimpairment		ST abROA		LT abROA		Completion	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Predicted based on CAR	26.039		-0.855			6.573		1.090**
	(30.177)		(1.031)			(4.985)		(0.404)
Predicted based on characteristics		0.708***		1.338***		1.188***		0.995***
		(0.138)		(0.368)		(0.228)		(0.059)
Observations	2,862	2,862	14,358	14,358	10,713	10,713	18,014	18,014
Adjusted R^2	0.000	0.027	0.000	0.039	0.000	0.041	0.000	0.148
Panel B. Is CAR Correlated with the "Predictable" Component of Outcomes?								
Dependent variable:	Predicted Outcome by a Characteristics Model							
	Nonimpairment		ST abROA		LT abROA		Completion	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CAR[-1, 1]	-0.104***		-0.014***			-0.010***		-0.008
	(0.005)		(0.001)		(0.003)		(0.017)	
CAR[-5, 5]		-0.078***		-0.009***		-0.008***		-0.011
		(0.009)		(0.002)		(0.003)		(0.008)
Observations	2,862	2,862	14,358	14,358	10,713	10,713	18,014	18,014
Adjusted R^2	0.010	0.009	0.005	0.006	0.001	0.002	0.000	0.000

(b) Table III — Out-of-Sample: Predicted versus Realized Outcomes

Reading notes. Figure 3 shows that CAR's weak performance is not limited to one decade. Table III is central: the characteristics-only model predicts realized outcomes out-of-sample, but CAR-only predictions do not. Panel B also shows that CAR is not correlated with the predictable component of outcomes generated from characteristics.

10.4 Figure 4 — Out-of-sample: Predicted versus realized outcomes; Table IV — Long-Term Returns and Predicted Outcomes

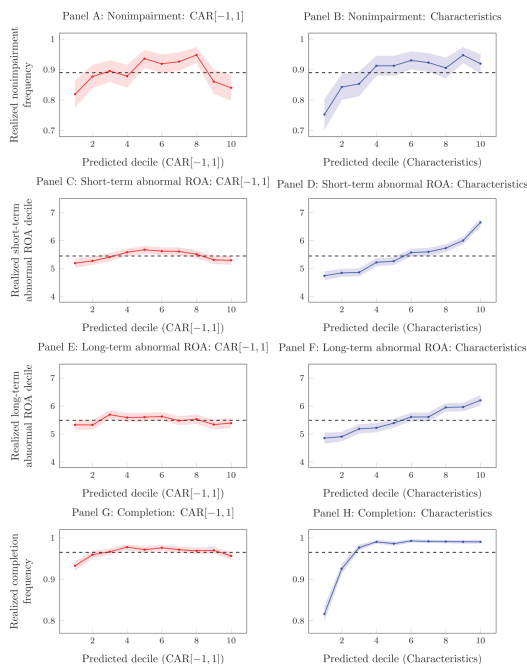


Figure 4. Out-of-sample: Predicted versus realized outcomes. This figure depicts out-of-sample results. We use the first half of each sample to fit OLS regressions that use nonimpairment, short- and long-term abnormal ROA, and deal completion as outcome variables. Panels A, C, E, and G include only acquirer CAR[-1, 1] as the independent variable. Panels B, D, F, and H include only deal and firm characteristics as the independent variables. Using the estimates, we obtain predicted outcome deciles for the second half of each sample. For our transaction-level measures, for each predicted probability decile we report the fraction of transactions with realized success or completion. We report the realized outcome decile for our firm-level measures for each predicted decile. The dashed line indicates the unconditional realized frequency (for success and completion) and the unconditional realized outcome decile (for short- and long-term abnormal ROA) for the second half of each sample. The shaded area represents the 95% confidence interval. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 4 — Out-of-sample: Predicted versus realized outcomes

Table IV
Long-Term Returns and Predicted Outcomes

This table reports 60-month equal-weighted DGTW-adjusted returns computed beginning at the month-end of the deal announcement date. In columns (1) to (4), we estimate yearly OLS regressions of the probability of nonimpairment, short- and long-term abnormal ROA, and the probability of completion, respectively, on firm and deal characteristics. We then compute the imputed outcome for each year and sort predicted values into 10 outcome deciles. We report the equal-weighted 60-month DGTW-adjusted BHAR for acquirers in the bottom-three and top-three deciles and the *p*-value for the difference test between the top and bottom deciles. The characteristics include log market capitalization, leverage and free cash flow scaled by lagged assets, Tobin's Q, previous-quarter market-adjusted stock returns, relative deal size, and a set of dummy variables for stock-only, mixed-payment, diversifying, contested, hostile, and public target deals. We measure firm-level characteristics in the year before the deal announcement. Columns (5) to (8) are computed analogously, except we use CAR[-1, 1] to predict outcomes.

Prediction Variable:	Characteristics-Only Model				CAR-Only Model			
	Abnormal ROA			Completion	Abnormal ROA		Completion	
	Nonimpair.	ST	LT		Nonimpair.	ST		LT
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Top three deciles	-2.7%	-5.1%	-3.8%	-5.0%	-6.9%	-10.9%	-8.9%	-10.9%
Bottom three deciles	-12.0%	-15.8%	-13.2%	-12.9%	-6.9%	-13.7%	-8.1%	-13.7%
Difference	9.2%	10.7%	9.4%	7.9%	0.1%	2.8%	-0.8%	2.8%
<i>p</i> -Value	0.005	0.000	0.000	0.000	0.886	0.158	0.726	0.089

(b) Table IV — Long-Term Returns and Predicted Outcomes

Reading notes. Figure 4 visually contrasts CAR-only and characteristics-only forecasts. The CAR panels are flat or nonmonotonic; the characteristics panels slope in the expected direction. Table IV validates the characteristics model using long-run returns: characteristics-based predicted outcomes generate large five-year return spreads, while CAR-based predicted outcomes do not.

10.5 Table V — “Listening” to CAR; Figure 5 — Correlation of CAR and outcomes with characteristics

Table V
“Listening” to CAR

This table reports 60-month equal-weighted DGTW-adjusted portfolio returns computed starting from the month-end of the deal announcement date. In columns (1) and (2), we estimate OLS regressions of the withdrawal probability on deal characteristics using the early years of the sample, before 2000. We then compute the imputed outcome for years after 2001. Columns (3) and (4) are computed analogously, except we use $CAR[-1, 1]$ to predict outcomes. In Panels A and B, we limit the sample to completed and withdrawn deals, respectively, and we sort predicted values into outcome deciles. We report the equal-weighted 60-month DGTW-adjusted mean (median) BHAR for acquirers in the bottom three and top three deciles. Panel C reports differences between the different signals and the overall return if one consistently “listened” to CAR.

Prediction Model:	60-Month DGTW-Adjusted BHAR			
	Characteristics		CAR	
	Mean	Median	Mean	Median
	(1)	(2)	(3)	(4)
Panel A: Completed Deals				
(a) Top three deciles of withdrawal prediction	-12%	-28%	-15%	-26%
(b) Bottom three deciles of withdrawal prediction	1%	-7%	-1%	-21%
Panel B: Withdrawn Deals				
(c) Top three deciles of withdrawal prediction	-9%	-25%	-35%	-50%
(d) Bottom three deciles of withdrawal prediction	-17%	-30%	-16%	-28%
Panel C: “Listening to CAR”				

(a) Table V — “Listening” to CAR

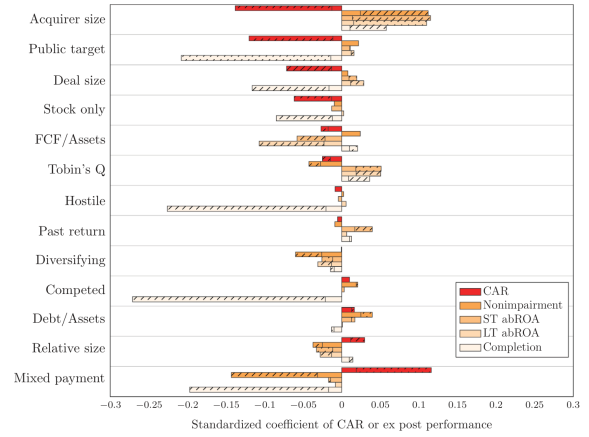


Figure 5. Correlation of CAR and outcomes with characteristics. This bar chart shows the standardized coefficients for regressions in which the dependent variable is CAR, nonimpairment, short- and long-term abnormal ROA, or completion on the various deal and firm characteristics. Each characteristic enters each regression individually (univariate regressions). The red bars indicate the standardized coefficients from regressions in which CAR is the dependent variable, and the four lighter bars indicate regressions for which nonimpairment, short- and long-term abnormal ROA, and completion are the dependent variables. The patterned portion of the bars indicates a coefficient larger than 1.96 standard errors of the standardized coefficient, that is, statistically significant at the 5% level or better. All acquirer characteristics are computed before the announcement. Leverage, free cash flows (FCF), assets, and Tobin's Q are computed the year before the announcement. Past returns are computed in the quarter before the announcement. (Color figure can be viewed at wileyonlinelibrary.com)

(b) Figure 5 — Correlation of CAR and outcomes with characteristics

Reading notes. Table V asks whether managers would have benefited from listening to CAR signals. The answer is no: consistently listening to CAR produces a negative return spread, while listening to the characteristics model produces a large positive spread. Figure 5 shows that the characteristics associated with high or low CAR often do not align with the characteristics associated with good or bad ex post outcomes.

10.6 Figure 6 — CAR-based predictors versus characteristics-based predictors; Table VI — Acquisition Outcomes and CAR, Grouped by Characteristics

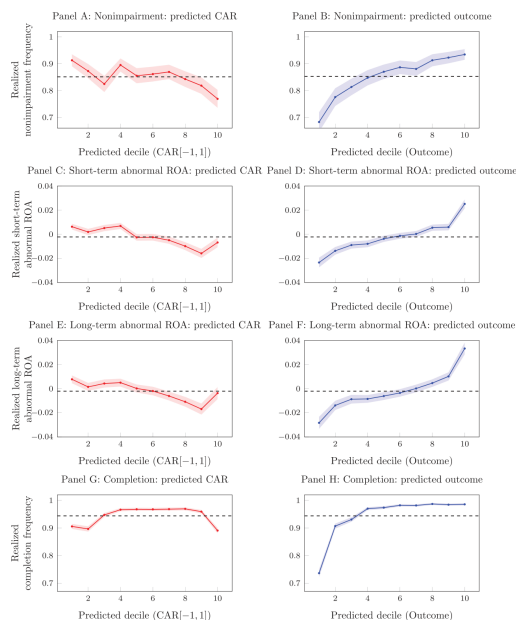


Figure 6. CAR-based predictors versus characteristics-based predictors. We use the coefficients from a regression of CAR on characteristics to obtain an in-sample predicted CAR for the sample of completed transactions, that is, a summary of what CAR would be given the set of deal and acquirer characteristics. We then sort the predicted CAR into deciles. On the left-hand side of the figure, for each predicted CAR decile, we report (solid red line) realized nonimpairment frequency (Panel A), average realized short- and long-term abnormal ROA (Panels C and E, respectively), and realized completion frequency (Panel G). Red shading indicates 95% confidence intervals. Similarly, we use the coefficients from regressions of ex post outcomes on characteristics to obtain in-sample predicted nonimpairment, short-term abnormal ROA, long-term abnormal ROA, and completion. We then sort predicted values into deciles, with results presented on the right-hand side of the figure. For each predicted decile, we report (solid blue line) realized nonimpairment frequency (Panel B), average realized short- and long-term abnormal ROA (Panels D and F, respectively), and realized completion frequency (Panel H). Blue shading indicates 95% confidence intervals.
(Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 6 — CAR-based predictors versus characteristics-based predictors

Table VI
Acquisition Outcomes and CAR, Grouped by Characteristics

This table reports the average of the acquisition outcome variables and CAR for acquisitions grouped by the characteristics identified in extant literature as correlated with CAR. *Rank* is the average rank of the four outcome variables. To facilitate interpretation, statistics within each column are color-coded from red (signifying the worst performance) to green (signifying the best performance) for each measure.

Acquisition Characteristics					Ex Post Outcomes					
Cash Only	Public Target	Large Acquirer	Large Relative Size	Avg CAR[-1, 1]	Non-impairment	ST abROA	LT abROA	Completion	Avg Rank	
Y			Y	0.028	0.753	-0.008	-0.008	0.935	14	
			Y	0.022	0.818	-0.007	-0.010	0.973	11	
		Y	Y	0.018	0.834	-0.006	-0.005	0.950	7	
Y	Y		Y	0.018	0.791	-0.015	-0.010	0.818	16	
Y		Y	Y	0.014	0.856	0.002	0.002	0.975	5	
Y				0.010	0.874	-0.008	-0.010	0.989	6	
Y	Y			0.008	0.925	-0.002	-0.013	0.876	10	
Y	Y	Y	Y	0.007	0.823	0.002	0.005	0.804	8	
				0.004	0.821	-0.013	-0.012	0.981	12	
Y		Y	Y	0.002	0.913	0.012	0.011	0.986	1	
		Y	Y	0.002	0.919	0.004	0.004	0.979	3	
Y	Y	Y		0.001	0.888	0.020	0.018	0.931	4	
		Y	Y	-0.003	0.953	0.008	0.009	0.938	2	
				-0.004	0.949	-0.014	-0.024	0.892	12	
			Y	-0.011	0.849	-0.012	-0.011	0.789	15	
	Y	Y	Y	-0.026	0.837	0.000	0.004	0.843	8	

(b) Table VI — Acquisition Outcomes and CAR, Grouped by Characteristics

Reading notes. Figure 6 turns CAR-based inferences into a single predictor and shows that it does not line up with realized outcomes. The characteristics-based predictors do. Table VI is the most intuitive example: common deal categories that appear best by CAR are not the categories with the best realized outcomes.

10.7 Figure 7 — Cisco's acquisitions and CARs; Figure 8 — \$CAR and \$Deal size

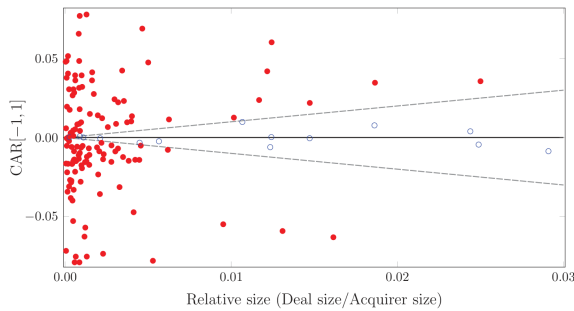


Figure 7. Cisco's acquisitions and CARs. This figure plots $CAR[-1, 1]$ and relative size (i.e., deal size divided by acquirer size) of Cisco's acquisitions within the range of relative size $\leq 3\%$ and CAR between $(\pm 8\%)$. The dashed gray lines represent the reasonable bounds on NPV (i.e., $|NPV/DealSize| \leq 1$). The blue hollow dots represent deals that fall within these reasonable bounds, and the red solid dots indicate CAR values outside these bounds. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 7 — Cisco's acquisitions and CARs

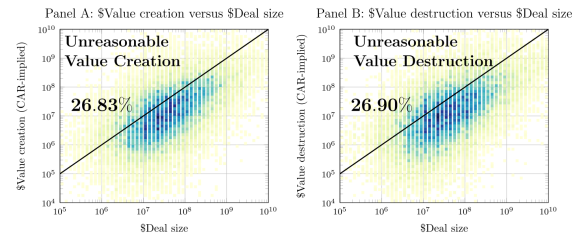


Figure 8. \$CAR and \$Deal size. The panels plot the frequency of transactions, presented as a function of \$CAR and \$Deal size. Panel A focuses on the subsample of positive \$CAR deals, and Panel B on the subsample of negative \$CAR deals. Darker colors represent a higher concentration of deals. The black lines represent $\$CAR = \$DealSize$. (Color figure can be viewed at wileyonlinelibrary.com)

(b) Figure 8 — \$CAR and \$Deal size

Reading notes. Figure 7 uses Cisco to show that CAR frequently implies economically implausible values relative to deal size. Figure 8 generalizes this idea: in a large share of positive- and negative- CAR deals, the implied value created or destroyed exceeds the deal size, which is difficult to reconcile with CAR as pure deal NPV .

10.8 Figure 9 — CAR around withdrawal announcements; Figure 10 — Std(\$CAR) and acquirer market capitalization and deal size

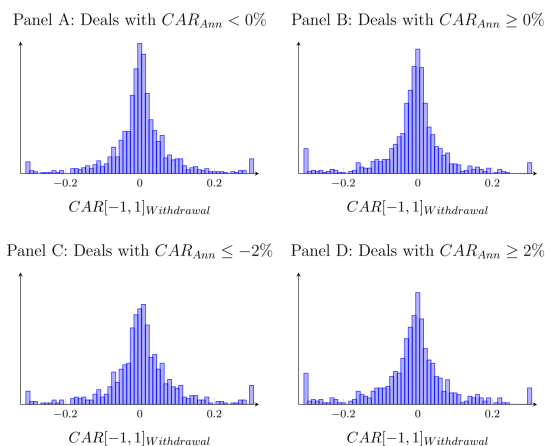


Figure 9. CAR around withdrawal announcements. This figure plots histograms of CAR around withdrawal announcements. Panels A and B present histograms of $CAR[-1, 1]_{Withdrawal}$ (withdrawal abnormal announcement returns) for negative and positive announcement CAR deals, respectively. Panels C and D present histograms of $CAR[-1, 1]_{Withdrawal}$ for deals with $CAR_{Announcement} \leq -2\%$ and $CAR_{Announcement} \geq 2\%$, respectively. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 9 — CAR around withdrawal announcements

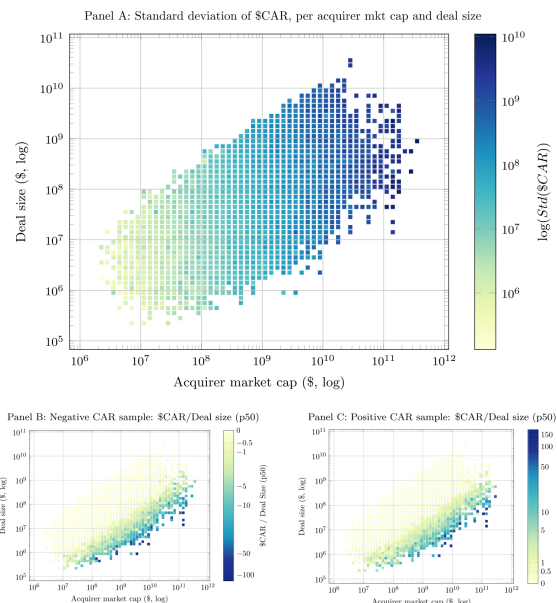


Figure 10. Std(\$CAR) and acquirer market capitalization and deal size. This figure presents heatmaps illustrating the variation in \$CAR based on acquirer market capitalization and deal size. We log-transform (base 10) both acquirer market capitalization and deal size, and then divide each dimension into bins with increments of 0.1. This creates a grid of bins based on the logged values, resulting in a final sample of 1,236 bins, where we restrict attention to those with more than five transactions. Panel A shows the standard deviation of \$CAR within each bin, with darker colors indicating higher variation. Panel B is limited to deals with negative CAR, and Panel C is limited to deals with positive CAR. In Panels B and C, cell colors represent the median value of \$CAR/Deal size on a $1 + \log(x)$ scale, and cells with median values $|x| > 1$ are represented by large markers. (Color figure can be viewed at wileyonlinelibrary.com)

(b) Figure 10 — Std(\$CAR) and acquirer market capitalization and deal size

Reading notes. If CAR were only deal NPV, withdrawal CAR should largely reverse announcement CAR. Figure 9 shows that this reversal is weak. Figure 10 shows the key scaling fact: variation in \$CAR moves with acquirer size far more than deal size, consistent with the acquirer-specific signal X dominating the deal signal.

10.9 Table VII — Determinants of the Variability in CAR; Figure 11 — Variability in CAR, by relative size

Table VII
Determinants of the Variability in CAR

This table reports standardized regression results of the variability in SCAR on deal size and the acquirer's market capitalization. Columns (1) to (3) present regressions using a sample based on individual acquisitions. The dependent variable is the logged absolute value of SCAR . SCAR is the change in the acquirer's market capitalization around the announcement event (window of $[-1, 1]$). In columns (4) to (7), observations are defined over bins of acquisition announcements, and the dependent variables are the logarithms of the standard deviation of acquisitions' SCAR within each bin. In columns (4) and (5), binning is based on the interaction of rounded logged deal size and rounded logged acquirer market capitalization (both rounded to the nearest 0.1). In columns (6) and (7), acquisitions are binned by acquirers. We require that bins have at least five acquisitions. All regressions include an intercept, which is not reported. We include the following deal characteristic controls (medians within each bin): leverage and free cash flow scaled by lagged assets, Tobin's Q , and previous-quarter market-adjusted stock returns, as well as dummy variables for stock-only, mixed-payment, diversifying, contested, hostile, and public target deals. We measure firm-level characteristics in the year before deal announcement. All variables are standardized to have a standard deviation of one. Standard errors are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Sample:	log(abs(SCAR))			log(std(SCAR))	
	log(abs(SCAR))			log(std(SCAR))	
	(1)	(2)	(3)	(4)	(5)
	All			Bins Based on AcqMktCap & DealSize	
log(Deal size)	0.106*** (0.010)	0.120*** (0.009)	0.123*** (0.007)	0.109*** (0.007)	0.055*** (0.012)
log(Acquirer market cap)	0.757*** (0.015)	0.742*** (0.012)	0.803*** (0.021)	0.882*** (0.006)	0.822*** (0.017)
Acquirer PE	No	No	Yes	No	No
Deal characteristic controls	No	Yes	No	No	No
Observations	39,585	39,585	39,687	1,296	2,509
Adjusted R^2	0.684	0.694	0.717	0.973	0.740

(a) Table VII — Determinants of the Variability in CAR

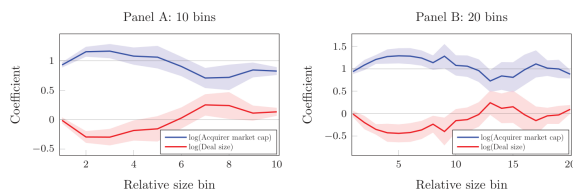


Figure 11. Variability in CAR, by relative size. This figure plots the estimated coefficients from equation (3) across relative size deciles. Panels A and B show results for specifications with 10 and 20 relative size bins, respectively. Blue (upper) lines represent estimates for the acquirer-size component (γ_1), and red lines (lower) represent estimates for the deal-size component (β_k). The shaded areas reflect 95% confidence intervals. These plots illustrate how the importance of deal size relative to acquirer size evolves with the relative size of the deal.
(Color figure can be viewed at wileyonlinelibrary.com)

(b) Figure 11 — Variability in CAR, by relative size

Reading notes. Table VII quantifies the scaling result: acquirer size explains $\$CAR$ variation much more strongly than deal size. Figure 11 shows that deal size becomes more informative only for very large relative-size deals, but even then it does not dominate the acquirer-size component.

11 One-page takeaway

- **Question:** Does acquirer CAR measure expected acquisition value creation?
- **Answer:** Generally no.
- **Evidence:** CAR does not meaningfully predict goodwill nonimpairment, abnormal ROA, or long-run returns. It predicts completion only weakly.
- **Better benchmark:** Standard deal and acquirer characteristics known at announcement predict outcomes and long-run returns more effectively.
- **Why CAR fails:** Acquisition announcements reveal acquirer-specific information as well as deal information. The acquirer-specific component dominates CAR .
- **Research implication:** Avoid using CAR as a stand-alone measure of acquisition quality or deal NPV . Use multiple ex post outcomes, characteristics-based predictions, and direct transaction-level measures when possible.